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Intestinal Nutrient Transporter Mechanisms

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Explain the apical and basolateral mechanisms of monosaccharide absorption and their electrochemical driving forces.
2. Explain how pancreatic and brush border enzymes generate absorbable peptides and distinguish amino acid transport types.
3. Explain inte[illegible]chanisms and predict the effects of transport disruptions.
4. Compare absorption mechanisms of fat-soluble vitamins and vitamin B12 and predict consequences of malabsorption.
5. Explain intestinal iron absorption including roles of DMT1, ferroportin, and hepcidin regulation.

CASE CONNECTION

JS, a 27-year-old male, presents to the clinic with chronic diarrhea and weight loss over the past four months. His labs reveal low serum iron, low vitamin B12, and mild hyponatremia.

What transporter disruptions could explain JS's symptoms? How do specific ions and nutrient absorption link to diarrhea? Consider your answers as you read, and we'll revisit JS at the end of the brick.

GO TO CONCLUSION ↓

Monosaccharide Absorption and Electrochemical Driving Forces (LO1)

Carbohydrate digestion ultimately yields monosaccharides—primarily glucose, galactose, and fructose—that must cross the intestinal epithelium to reach the bloodstream. This process relies on a coordinated set of membrane transporters expressed on the apical (luminal) and basolateral (serosal) surfaces of enterocyt[illegible]handles which sugar, and what drives each transport event, is fundamental to predicting how dietary carbohydrates are absorbed and what happens when these systems fail.

Apical and Basolateral Transporters

Apical uptake. Two distinct transporters handle monosaccharides at the apical membrane. **Sodium-glucose linked transporter 1 (SGLT1)** is a secondary active transporter that couples the movement of two Na^+ ions to each glucose or galactose molecule. The driving force is the inward Na^+ electrochemical gradient, maintained by the basolateral **Na^+/K^+-ATPase** (3 Na^+ out, 2 K^+ in), which keeps intracellular Na^+ low. This arrangement allows SGLT1 to concentrate glucose and galactose inside the enterocyte even against their concentration gradients — without directly hydrolyzing ATP. **Glucose transporter 5 (GLUT5)**, in contrast, is a facilitative transporter that absorbs fructose down its concentration gradient, with no ion coupling.

Basolateral export. All three monosaccharides exit via **glucose transporter 2 (GLUT2)**, a high-capacity, low-affinity facilitative transporter on the basolateral membrane. Because active apical uptake raises intracellular sugar concentrations, GLUT2 moves them down a favorable gradient into the portal circulation — sustained by continuous removal of sugars by portal blood flow.

Clinical implications. This architecture produces distinct malabsorption syndromes: SGLT1 defects impair glucose and galactose absorption specifically, while GLUT5 defects selectively impair fructose absorption.

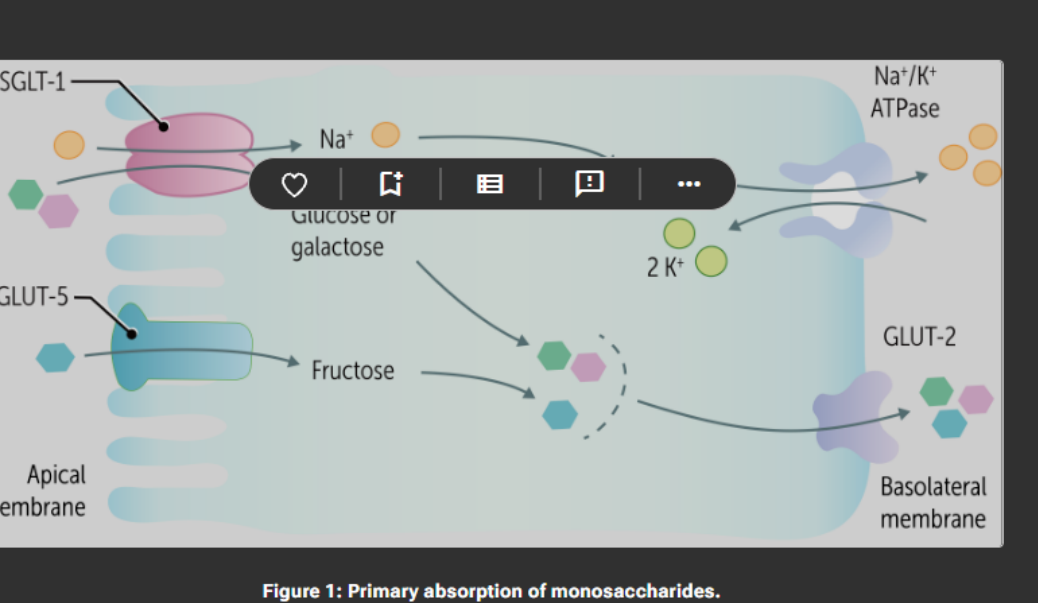

Figure 1: Primary absorption of monosaccharides.

What are the apical transporters for glucose, galactose, and fructose, and what drives each?

Protein Digestion and Amino Acid Transport (LO2)

Dietary proteins are l[illegible]be absorbed intact. Their digestion requires a sequential enzymatic process that begins in the stomach and continues in the small intestine, ultimately generating small peptides and free amino acids that can be transported across the enterocyte. The intestinal phase of this process involves two enzyme groups working in concert: pancreatic endopeptidases secreted into the lumen and brush border exopeptidases anchored to the apical membrane.

Enzyme Roles

Pancreatic endopeptidases—including trypsin, chymotrypsin, and elastase—cleave peptide bonds within protein chains, producing shorter oligopeptides. These enzymes are secreted as inactive zymogens and activated in the duodenum; trypsinogen is converted to trypsin by enteropeptidase on the brush border, and trypsin then activates the other zymogens. The result is a mixture of oligopeptides and some free amino acids entering the brush border zone.

Brush border exopeptidases—including aminopeptidases and dipeptidyl peptidases—cleave amino acids from the N-terminus of oligopeptides, generating free amino acids and di- and tripeptides. These small substrates are the final absorbable forms. Free amino acids are taken up by Na^+-dependent transporters, while di- and tripeptides are absorbed intact via the peptide transporter PepT1.

CLINICAL CORRELATION

Patients with brush border enzyme deficiencies accumulate oligopeptides that cannot be absorbed,[illegible]alnutrition.

Amino Acid vs. Peptide Transport

Free amino acid absorption occurs via multiple Na^+-dependent cotransporters on the apical membrane, each with specificity for different amino acid classes (neutral, basic, acidic, and imino acids). Like SGLT1, these transporters harness the inward Na^+ electrochemical gradient maintained by basolateral Na^+/K^+-ATPase. This secondary active mechanism allows amino acid uptake even against concentration gradients, ensuring efficient absorption even at low luminal concentrations.

PepT1, the H^+-dependent oligopeptide transporter, uses a different driving force. It couples di- and tripeptide uptake to the inward H^+ gradient generated by the Na^+/H^+ exchanger (NHE3) on the apical membrane. NHE3 extrudes H^+ into the unstirred water layer adjacent to the brush border, creating a local acidic microenvironment. PepT1 then uses this H^+ gradient to drive peptide uptake. Once inside the enterocyte, most di- and tripeptides are hydrolyzed by cytoplasmic peptidases to free amino acids, which then exit via basolateral transporters. PepT1's broad substrate specificity makes it a pharmacologically important transporter—many orally administered drugs, including certain antibiotics and antivirals, are absorbed via PepT1.

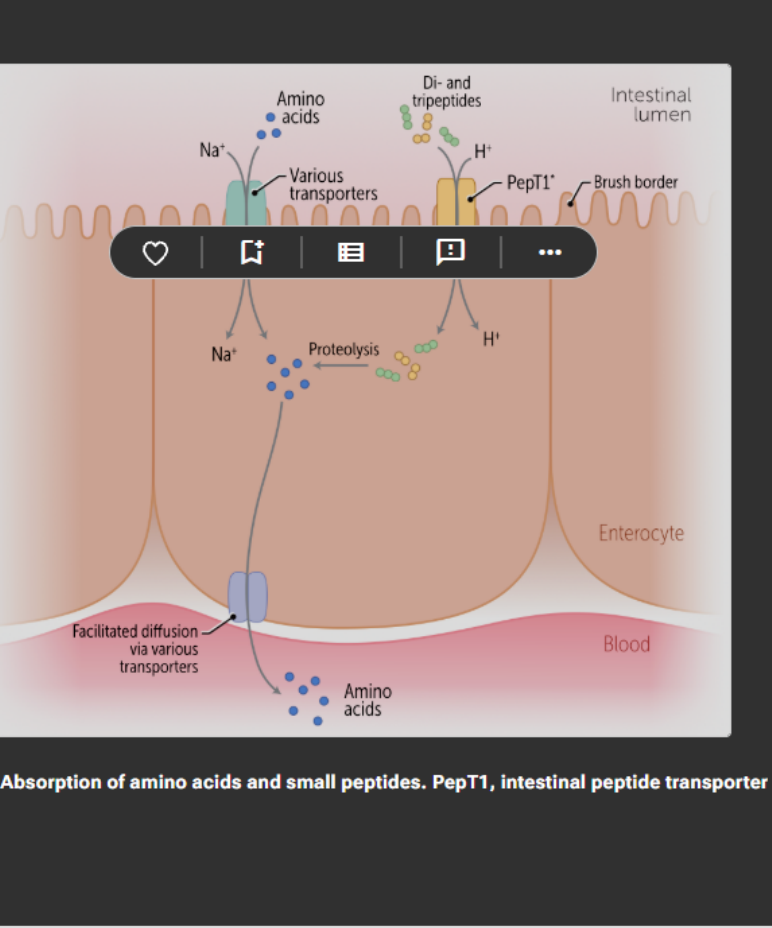

Figure 2: Absorption of amino acids and small peptides. PepT1, intestinal peptide transporter 1.

ACTIVE LEARNING QUESTION

Brush border aminopeptidases cleave amino acids from the N-terminus of luminal oligopeptides. In a patient who lacks functional brush border aminopeptidases, which substrate accumulates in the intestinal lumen, and how is peptide absorption affected?

- A. ○ Glucose accumulates; SGLT1 uptake rises
- B. ○ Monosaccharides accumulate; GLUT2 export falls
- C. ○ Oligo[illegible]id uptake fall
- D. ○ Free amino acids accumulate; PepT1 uptake rises
- E. ○ Dipeptides accumulate; NHE3 exchange fails

Explanation (requires correct answer)

Intestinal Na^+, Cl^-, and Water Absorption (LO3)

The small and large intestines absorb enormous volumes of fluid daily—roughly 8–9 liters enter the small intestine, and only about 100–200 mL is lost in stool. This remarkable efficiency depends on tightly regulated electrolyte transport, particularly of Na^+ and Cl^-, which osmotically drive water absorption. Disruptions in these transport mechanisms are the primary cause of diarrheal disease, making this physiology directly relevant to clinical practice.

Na^+, Cl^-, and K^+ Transporters

Na^+ absorption occurs via several mechanisms depending on intestinal segment. In the small intestine, Na^+ is co-absorbed with nutrients (glucose via SGLT1, amino acids via Na^+-dependent transporters) and also via electroneutral NHE3, which exchanges luminal Na^+ for intracellular H^+. In the colon, epithelial Na^+ channels (ENaC) mediate electrogenic Na^+ absorption, regulated by aldosterone. Basolateral Na^+/K^+-ATPase drives all these processes by maintaining low intracellular Na^+.

Cl^- absorption in the [illegible]orption via the paired action of NHE3 and the Cl^-/HCO_3^- exchanger (DRA, also called SLC26A3). Intracellular carbonic anhydrase generates H^+ and HCO_3^- from CO_2 and water; NHE3 extrudes the H^+ in exchange for luminal Na^+, while DRA exports the HCO_3^- in exchange for luminal Cl^-. The net result is electroneutral NaCl absorption, the dominant mechanism in the jejunum and ileum.

Potassium absorption in the small intestine occurs primarily by passive paracellular diffusion, driven by solvent drag as water follows absorbed solutes into the interstitium. Unlike Na^+ and Cl^-, K^+ does not rely on dedicated apical transporters in the small intestine. In the colon, K^+ handling is bidirectional: absorption occurs via an apical H^+/K^+-ATPase, while secretion occurs through apical K^+ channels — with the net direction influenced by aldosterone, which promotes K^+ secretion.

Water Movement

Water absorption is entirely passive and follows the osmotic gradient created by solute absorption. As Na^+, Cl^-, and nutrients are transported into the enterocyte and then into the lateral intercellular spaces, local osmolality rises, drawing water through aquaporin channels and tight junctions. Aquaporins 1, 3, 7, and 8 are expressed in intestinal epithelial cells and facilitate transcellular water movement. The paracellular route through tight junctions is particularly important in the small intestine, where junctions are relatively leaky. In the colon, tighter junctions mean water absorption is more dependent on active solute transport.

Diarrhea Types

Understanding trans[illegible]precise prediction of diarrhea mechanisms. Osmotic diarrhea occurs when unabsorbed, osmotically active solutes remain in the lumen, retaining water. This happens when a transporter or brush border enzyme responsible for nutrient handling fails — for example, lactase deficiency leaves undigested lactose in the small intestine, where it osmotically retains water; colonic fermentation of the unabsorbed lactose then adds gas and worsens the load. Osmotic diarrhea characteristically resolves with fasting, because removing the unabsorbed solute eliminates the osmotic load.

Secretory diarrhea, in contrast, results from active ion secretion or inhibition of absorption that persists even during fasting. Cholera toxin irreversibly activates adenylyl cyclase in enterocytes, raising cyclic AMP (cAMP), which activates CFTR and inhibits NHE3. The result is massive Cl^- secretion and Na^+ absorption failure, producing large-volume, watery stools. Hyponatremia can result from prolonged secretory diarrhea causing Na^+ depletion. This principle underlies oral rehydration therapy (see Clinical Correlation).

CLINICAL CORRELATION

Oral Rehydration Therapy

Cholera toxin activates CFTR and inhibits NHE3, producing massive secretory diarrhea, but **SGLT1 remains intact**. Oral rehydration solutions exploit this by combining glucose with sodium: SGLT1 couples glucose absorption to Na^+ uptake, and water follows osmotically. The same transporter that drives postprandial sugar absorption becomes the therapeutic bypass that restores fluid balance, the cornerstone of diarrheal disease management worldwide.

ACTIVE LEARNING QUESTION

A patient has profuse watery diarrhea that continues despite fasting. Which mechanism best explains this presentation?

- A. ○ Lactase deficiency
- B. ○ CFTR overactivation
- C. ○ SGLT1 deficiency
- D. ○ NHE3 loss
- E. ○ ENaC loss

Explanation (requires correct answer)

Fat-Soluble Vitamin and Vitamin B12 Absorption (LO4)

Vitamins are essential micronutrients with highly specialized absorption mechanisms. Fat-soluble vitamins (A, D, E, K) and water-soluble vitamin B12 are absorbed by fundamentally different pathways, reflecting their distinct physicochemical properties. Because these pathways depend on different anatomical sites and[illegible]ion produces predictable and clinically distinct deficiency syndromes.

Fat-Soluble Vitamins

Vitamins A, D, E, and K are lipophilic and require solubilization in bile salt micelles for intestinal absorption. Dietary fat digestion by pancreatic lipase releases these vitamins from food matrices, and they partition into mixed micelles along with fatty acids and monoglycerides. These micelles diffuse to the brush border, where the vitamins are released and passively diffuse across the apical membrane into the enterocyte. Inside the cell, fat-soluble vitamins are incorporated into chylomicrons along with triglycerides and exported via exocytosis into intestinal lymphatics (lacteals), bypassing the portal circulation.

CLINICAL CORRELATION

Because fat-soluble vitamin absorption depends on bile salts and pancreatic lipase, any condition impairing fat digestion or bile secretion—such as cholestatic liver disease, pancreatic exocrine insufficiency, or ileal disease disrupting bile salt reabsorption—can cause fat-soluble vitamin deficiency. The clinical consequences are well-defined: vitamin A deficiency causes night blindness and xerophthalmia; vitamin D deficiency leads to rickets or osteomalacia; vitamin E deficiency causes hemolytic anemia and neuropathy; and vitamin K deficiency impairs coagulation, manifesting as easy bruising and prolonged prothrombin time.

Vitamin B12 Pathw[illegible]

Most water-soluble vitamins — including vitamin C, thiamine (B1), riboflavin (B2), niacin (B3), pyridoxine (B6), and folate (B9) — are absorbed in the proximal small intestine via Na^+-dependent co-transporters or carrier-mediated facilitated diffusion. Because these transporters are expressed in the duodenum and jejunum, proximal mucosal disease can produce multiple water-soluble vitamin deficiencies simultaneously. Vitamin B12 is the major exception.

Vitamin B12 (cobalamin) absorption is a multi-step process requiring specific binding proteins. In the stomach, dietary B12 is released from food proteins by gastric acid and pepsin, then bound by haptocorrin (R-protein) secreted in saliva and gastric juice. In the duodenum, pancreatic proteases degrade haptocorrin, releasing B12 to bind intrinsic factor (IF), a glycoprotein secreted by gastric parietal cells. The B12-IF complex is highly resistant to digestion and travels to the terminal ileum, where it binds the cubilin-amnionless receptor complex on enterocytes. The complex is endocytosed, B12 is released intracellularly, and it exits the basolateral membrane bound to transcobalamin II for portal transport.

This elaborate pathway creates multiple potential failure points. Pernicious anemia results from autoimmune destruction of parietal cells, eliminating IF production. Gastrectomy removes the source of both gastric acid and IF. Terminal ileal disease (e.g., Crohn's disease) or resection destroys the absorption site. Pancreatic exocrine insufficiency prevents haptocorrin degradation, trapping B12 in an unabsorbable form. In all these scenarios, B12 deficiency develops over months to years due to large hepatic stores, ultimately causing megaloblastic anemia and subacute combined degeneration of the spinal cord.

Malabsorption Effects

The distinct absorption pathways of fat-soluble vitamins and B12 mean that different diseases produce different deficiency patterns. Ileal disease selectively impairs B12 absorption (and bile salt recirculation, secondarily affecting fat-soluble vitamins). Cholestatic disease primarily impairs fat-soluble vitamin absorption. Pernicious anemia selectively depletes B12. Recognizing these patterns allows clinicians to predict which vitamins are at risk based on the underlying pathology, guiding targeted supplementation strategies.

How do fat-soluble vitamins and vitamin B12 differ in their intestinal absorption mechanisms?

Iron Absorption: DMT1, Ferroportin, and Hepcidin (LO5)

Iron is an essential mineral for oxygen transport, DNA synthesis, and cellular respiration, yet it is also potentially toxic in excess. The body has no regulated excretory pathway for iron, so absorption is the primary control point for iron homeostasis. The duodenum and proximal jejunum are the principal sites of iron absorption, and the process involves distinct apical uptake and basolateral export steps, each subject to systemic regulation.

DMT1 Transport

Dietary iron exists in two forms: heme iron (from animal sources) and non-heme iron (from plant sources). Non-heme iron is predominantly in the ferric (Fe^{3+}) form and must be reduced to ferrous iron (Fe^{2+}) by the brush border enzyme duodenal cytochrome b (Dcytb) before absorption. Divalent metal transporter 1 (DMT1) on the apical membrane then transports Fe^{2+} into the enterocyte. DMT1 is a H^+-coupled transporter, using the acidic microenvironment of the duodenum to drive Fe^{2+} uptake. Heme iron is absorbed via a separate, less well-characterized pathway involving endocytosis of the heme-receptor complex, with intracellular heme oxygenase releasing Fe^{2+}.

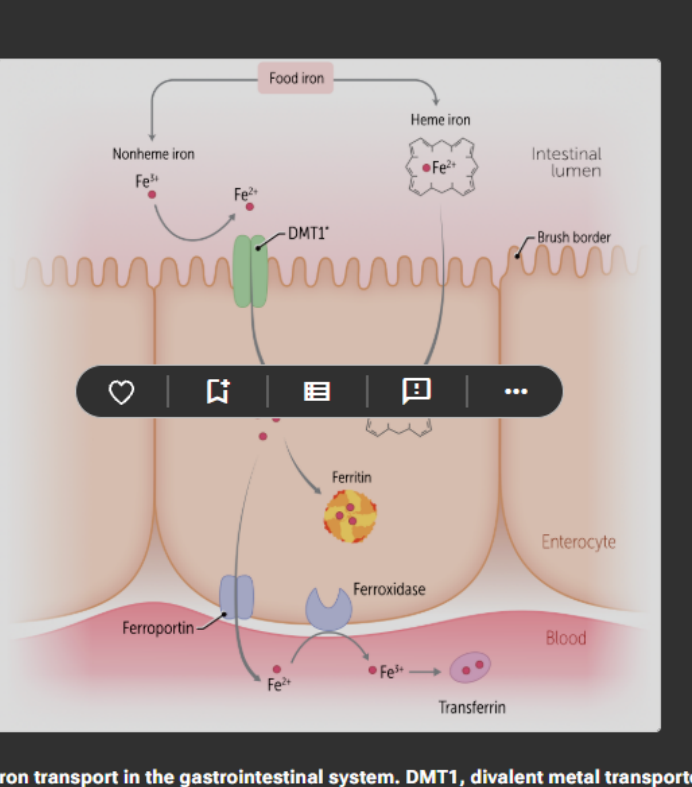

Figure 3: Iron transport in the gastrointestinal system. DMT1, divalent metal transporter 1.

Ferroportin Export

Once inside the enterocyte, Fe^{2+} can be stored bound to ferritin or exported across the basolateral membrane. Ferroportin (FPN1) is the only known cellular iron exporter in mammals and is essential for basolateral iron transfer into the portal circulation. As Fe^{2+} exits via ferroportin, it is oxidized to Fe^{3+} by hephaestin (a membrane-bound ferroxidase), allowing binding to transferrin in the blood for systemic distribution. If iron is not exported and remains stored in ferritin within the enterocyte, it is lost when the cell is shed at the end of its lifespan—a mechanism that limits iron absorption when body stores are adequate.

Hepcidin Regulation

Hepcidin, a peptide hormone produced by the liver, is the master regulator of systemic iron homeostasis. When body iron stores are replete or during inflammation, hepcidin levels rise. Hepcidin binds to ferroportin on enterocytes (and mac[illegible]on and degradation. This blocks basolateral iron export, trapping iron in enterocytes and reducing absorption. Conversely, when iron stores are low or erythropoiesis is increased, hepcidin levels fall, ferroportin expression increases, and iron absorption rises.

Dysregulation of the hepcidin-ferroportin axis underlies several important diseases, producing opposite clinical pictures depending on the direction of disruption (see Clinical Correlation).

CLINICAL CORRELATION

Hepcidin-Ferroportin Dysregulation

The hepcidin-ferroportin axis produces opposite clinical pictures depending on the direction of dysregulation. In hereditary hemochromatosis, an autosomal recessive disorder caused by HFE mutations, inappropriately low hepcidin allows unchecked ferroportin activity. Iron overload follows, producing the classic triad of **cirrhosis, diabetes mellitus, and skin hyperpigmentation** ("bronze diabetes"), with **dilated cardiomyopathy** a common additional complication. Diagnosis: elevated transferrin saturation and ferritin; confirmed by HFE genotyping.

In **anemia of chronic disease**, inflammatory cytokines — particularly **IL-6** — drive hepcidin overproduction. Ferroportin is degraded, trapping iron in macrophages and enterocytes despite adequate stores. Labs show low serum iron with normal or elevated ferritin, distinguishing it from iron deficiency anemia where ferritin is low.

ACTIVE LEARNING QUE[illegible]

A patient with chronic inflammation has low serum iron but normal ferritin. Which mechanism best explains this finding?

- A. ○ DMT1 upregulation
- B. ○ Elevated hepcidin degrading ferroportin
- C. ○ Dcytb deficiency
- D. ○ Transferrin deficiency
- E. ○ Low hepcidin levels

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JS, what transporter disruptions could explain his diarrhea, iron deficiency, and vitamin B12 deficiency? How do impaired ion and nutrient absorption link to his symptoms?

JS's presentation is consistent with diffuse small intestinal disease affecting multiple absorptive sites. Crohn's disease with both proximal and distal involvement could explain all findings. Damage to the proximal brush border would impair SGLT1 and Na^+-dependent transporters, reducing nutrient-coupled Na^+ absorption and caus[illegible]ars and amino acids. Sodium loss fr[illegible]yponatremia. Iron deficiency reflects impaired DMT1-mediated uptake in the damaged duodenal mucosa. B12 deficiency is most consistent with terminal ileal involvement, where the cubilin receptor mediates B12-IF complex uptake. Celiac disease could account for the iron deficiency and osmotic diarrhea through proximal mucosal damage, but because it spares the terminal ileum, it would not typically explain the B12 deficiency. This distinction helps narrow the differential toward a process with distal involvement such as Crohn's. Notably, the relatively rapid four-month onset suggests preexisting marginal B12 stores, since malabsorption alone typically takes longer to deplete hepatic reserves. Together, these findings map onto the transporter mechanisms covered in this brick.

Summary

- Monosaccharide absorption relies on SGLT1 (Na^+-coupled glucose/galactose uptake), GLUT5 (facilitated fructose uptake) at the apical membrane, and GLUT2 for basolateral export; the Na^+ gradient maintained by Na^+/K^+-ATPase is the primary driving force.
- Pancreatic endopeptidases generate oligopeptides from dietary proteins, while brush border exopeptidases produce free amino acids and di/tripeptides; free amino acids are absorbed via Na^+-dependent transporters, and di/tripeptides via H^+-dependent PepT1.
- Intestinal Na^+ absorption occurs via SGLT1, NHE3, and ENaC; Cl^- absorption is coupled via DRA; water follows osmotically. Disruption causes osmotic diarrhea (unabsorbed solutes) or secretory diarrhea (active ion secretion, e.g., CFTR overactivation).
- Fat-soluble vitamins (A, D, E, K) require bile micelles and are absorbed passively, exiting[illegible]tamin B12 requires intrinsic factor and cubilin receptor-mediated endocytosis in the terminal ileum.
- Intestinal iron absorption involves apical Fe^{2+} uptake via DMT1 (H^+-coupled) after reduction by Dcytb, and basolateral export via ferroportin with oxidation by hephaestin; hepcidin regulates iron homeostasis by promoting ferroportin degradation.
- Selective malabsorption patterns reflect specific transporter or pathway failures: ileal disease impairs B12 and bile salt absorption; cholestasis impairs fat-soluble vitamins; duodenal mucosal damage impairs iron and monosaccharide absorption.
- Clinical presentations of transporter dysfunction—diarrhea type, specific nutrient deficiencies, electrolyte disturbances—can be predicted from knowledge of which transporters are affected and their anatomical locations.

Review Questions

1. Which apical transporter absorbs glucose and galactose using the Na^+ gradient?

- A. ○ SGLT1
- B. ○ GLUT5
- C. ○ GLUT2
- D. ○ Na^+/K^+-ATPase
- E. ○ PepT1

Explanation (requires correct answer)

2. Which transporter absorbs di- and tripeptides using the H^+ gradient?

- A. ○ Na^+-dependent transporters
- B. ○ Brush border aminopeptidases
- C. ○ SGLT1
- D. ○ NHE3 exchanger
- E. ○ PepT1

Explanation (requires correct answer)

3. Which ion transporter disruption most likely causes secretory diarrhea persisting during fasting?

- A. ○ Lactase deficiency
- B. ○ CFTR overactivation
- C. ○ SGLT1 de[illegible]
- D. ○ NHE3 loss
- E. ○ ENaC loss

Explanation (requires correct answer)

4. How is vitamin B12 absorbed differently from fat-soluble vitamins?

- A. ○ Passive diffusion via micelles
- B. ○ Direct uptake in duodenum
- C. ○ Incorporation into chylomicrons
- D. ○ Intrinsic factor and receptor-mediated

Explanation (requires correct answer)

5. What explains low serum iron with normal ferritin in chronic inflammation?

- A. ○ DMT1 upregulation
- B. ○ Elevated hepcidin degrading ferroportin
- C. ○ Dcytb de[illegible]
- D. ○ Transferrin deficiency
- E. ○ Low hepcidin levels

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ Explain the apical and basolateral mechanisms of monosaccharide absorption and their electrochemical driving forces.
- 02 ☐ Explain how pancreatic and brush border enzymes generate absorbable peptides and distinguish amino acid transport types.
- 03 ☐ Explain intestinal Na^+, Cl^-, and water absorption mechanisms and predict the effects of transport disruptions.
- 04 ☐ Compare absorption mechanisms of fat-soluble vitamins and vitamin B12 and predict consequences of malabsorption.
- 05 ☐ Explain [illegible] of DMT1,

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Objective Confidence: 0/5